Kenya Meteorological Department
REPUBLIC OF KENYA

Climate Hazards Center
UC SANTA BARBARA

KIT
Karlsruher Institut für Technologie

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Early March Sub-seasonal Forecasts indicate bountiful rains and a good Start of Season in Kenya

March 3rd, 2026

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- CHIRPS3
- SubC Forecasts
- System Integration

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MINISTRY OF ENVIRONMENT, CLIMATE CHANGE AND FORESTRY
KENYA METEOROLOGICAL DEPARTMENT
REPUBLIC OF KENYA

Improved Monthly and Sub-Seasonal Forecasts

Improved Agricultural Advisories

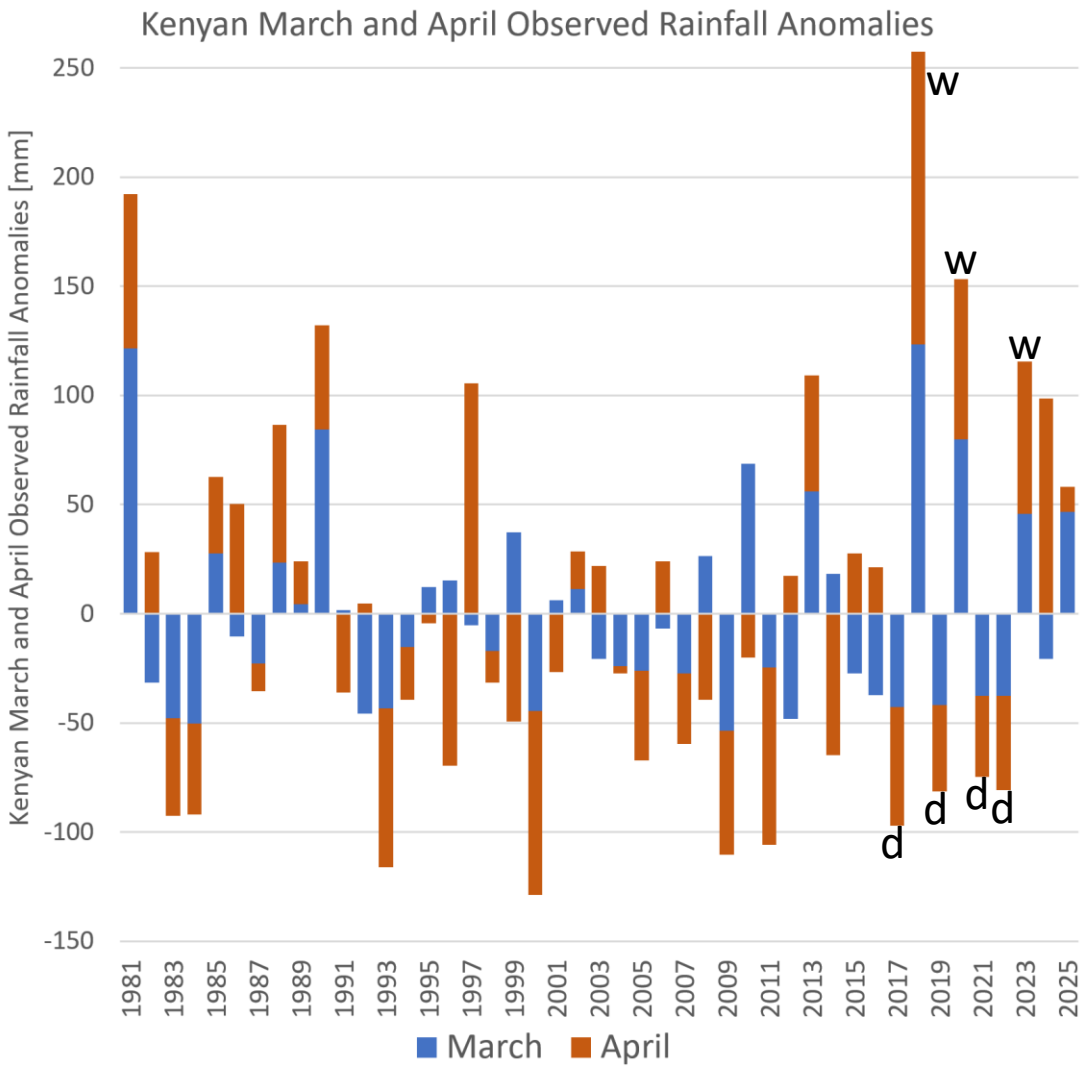
Improved Flood Alerts

Okello et al. (2026) How developing National Integrated Observations-Forecast Systems can build community resilience, GHACOF 72

Context – Frequent Extreme March and April Rains provide Kenyan farmers opportunity and peril

- Since 2017, Kenya has seen four seasons in which both March and April was very dry (dry seasons in 2017, 2019, 2021 and 2022), and four seasons in which both March and April were above normal (wet seasons in 2018, 2020, 2023 and 2025).

- We will see in this presentation that the good growing seasons in 2018, 2020 and 2023 **were predictable, in early March.**

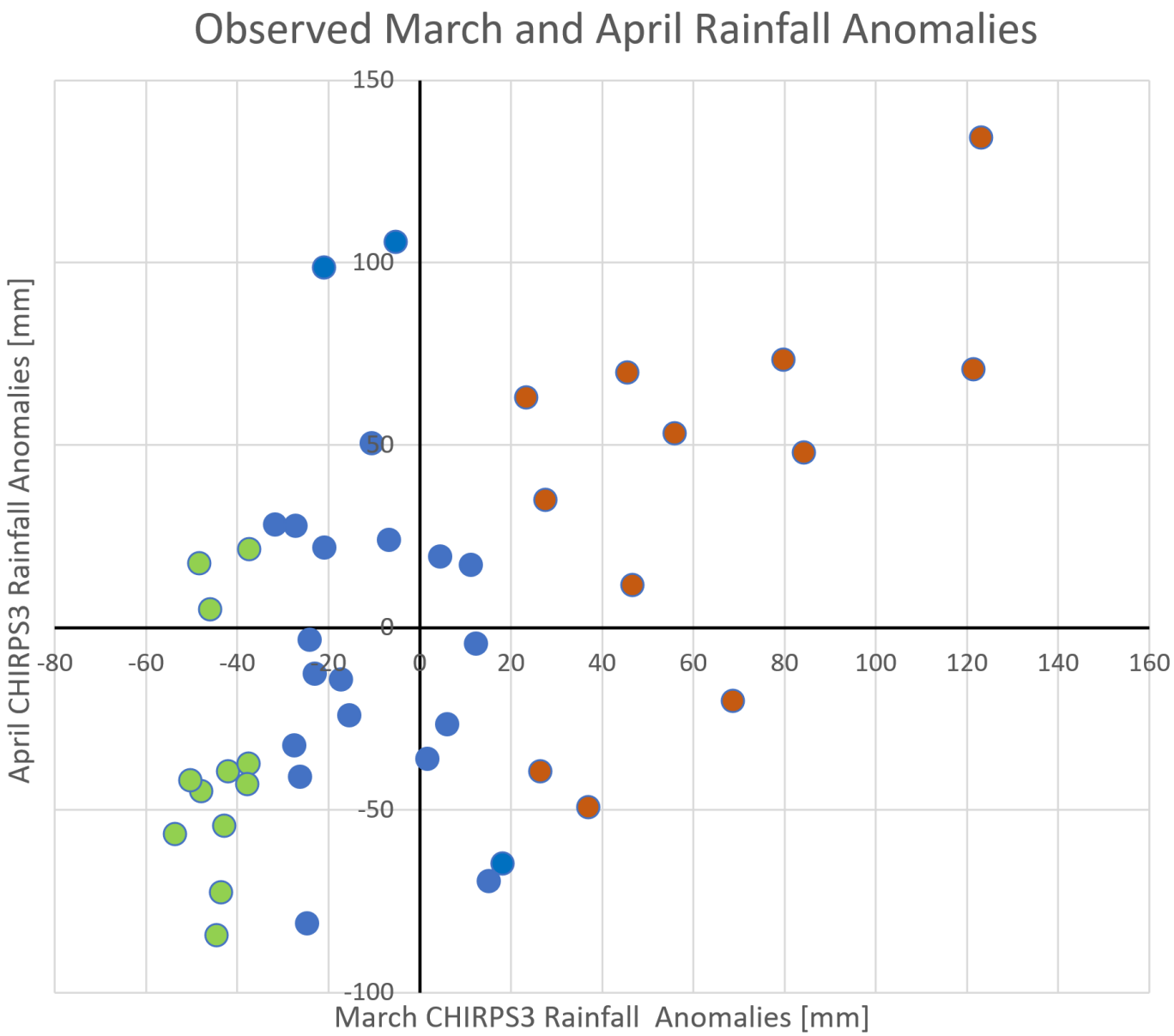


Context – Frequent Extreme March and April Rains provide Kenyan farmers opportunity and peril

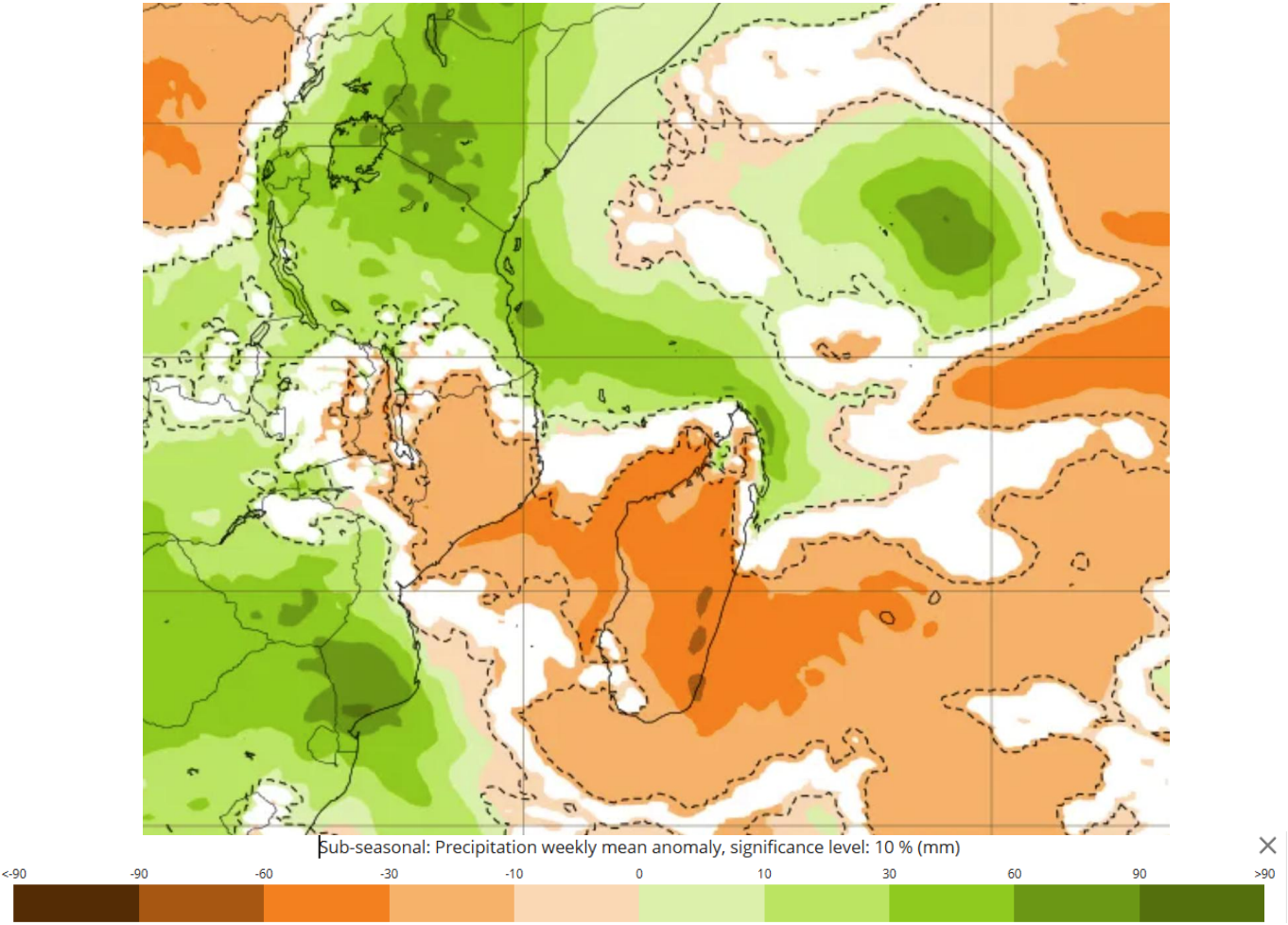
- Looking at a scatterplot of March and April Kenya rainfall anomalies, we can see that when March rainfall anomalies fall in the bottom quartile (green circles), we find a substantial downward shift in the April anomalies, with 75% of the April anomalies less than zero.

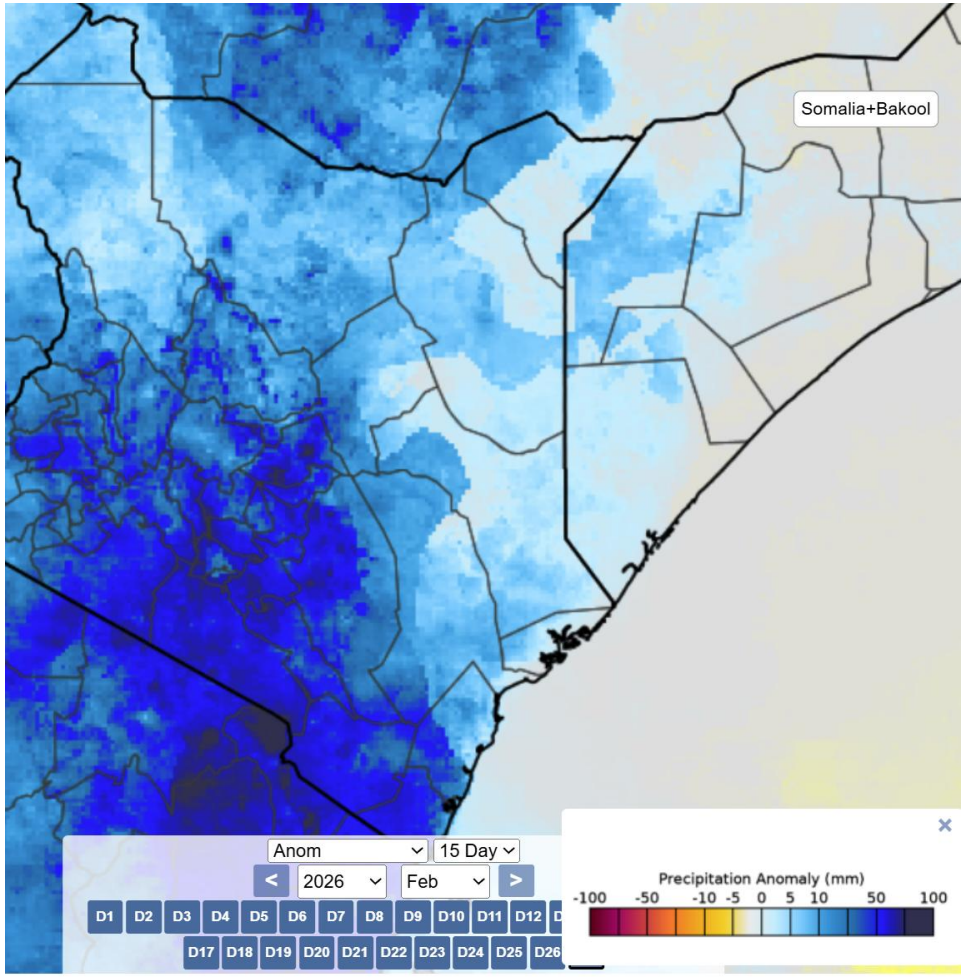
- An examination of the months with upper-quartile March rains (red circles) indicates an opposite shift, with 75% of the April anomalies above zero.

- These results suggest that March forecasts can help us identify high-risk and low-risk planting conditions.



Current forecasts suggest that early March rains are likely to be very exceptional over western Kenya

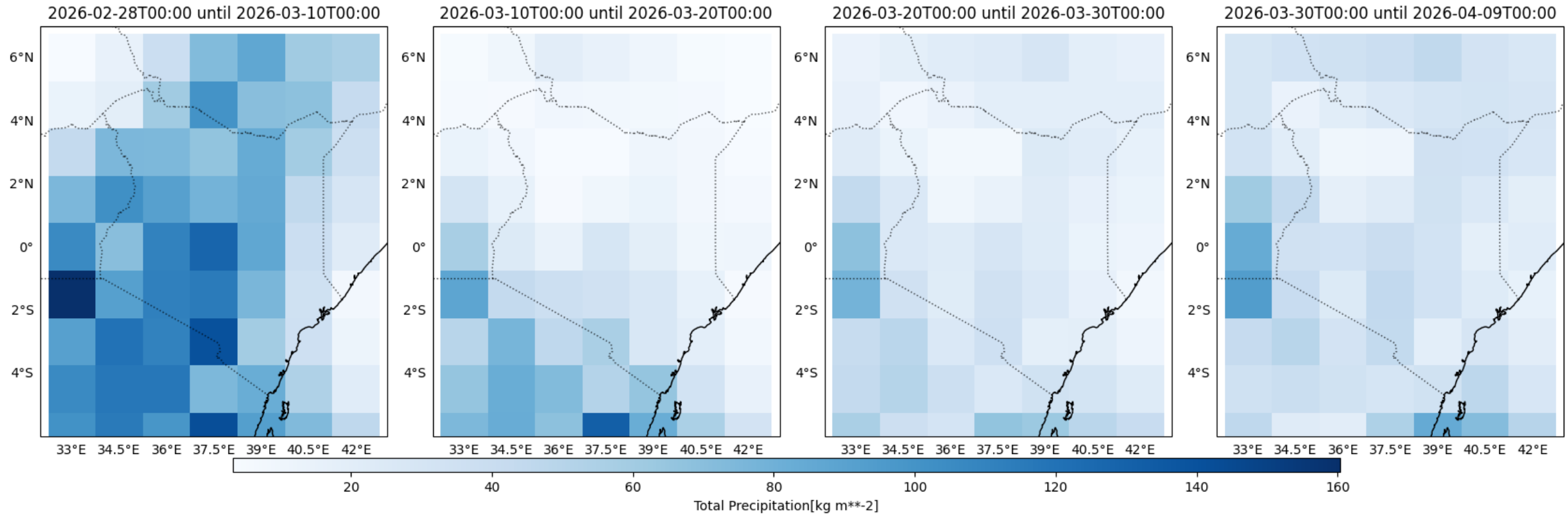




March 2nd to 9th ECMWF sub-seasonal rainfall
Forecasts anomalies
Accessed March 2nd

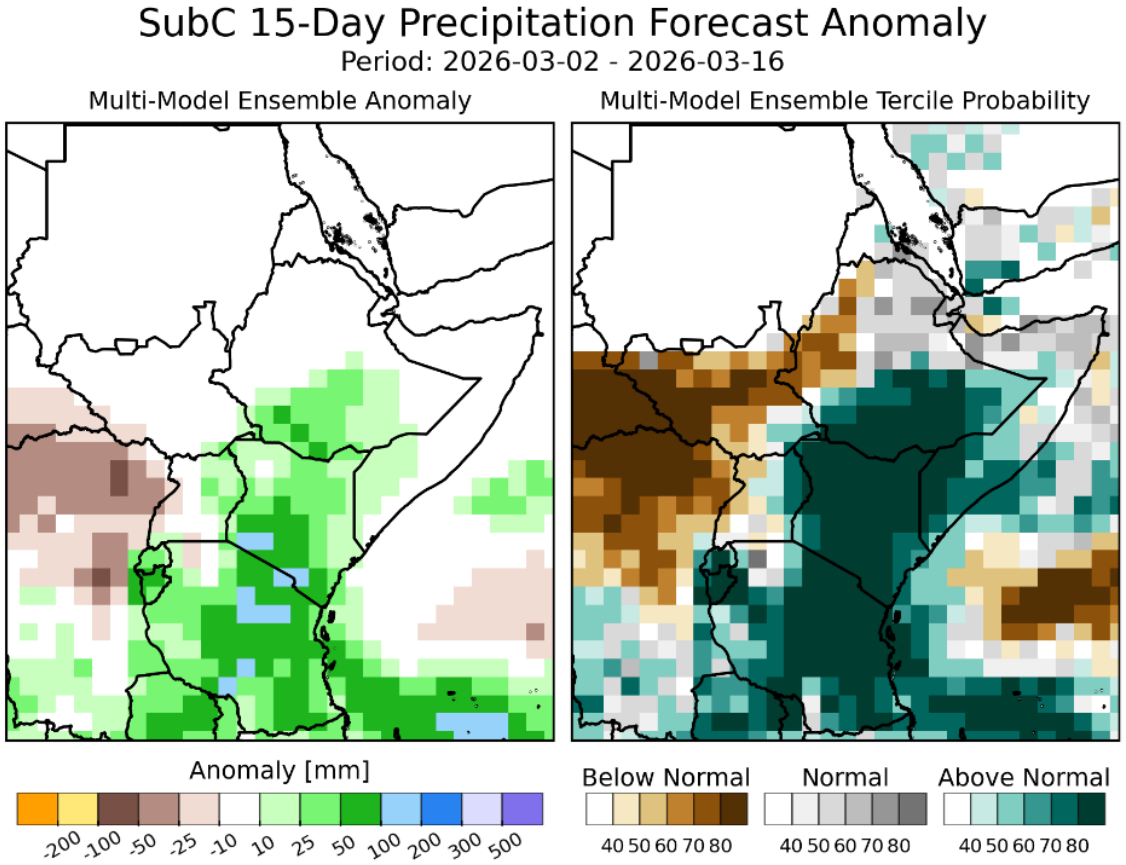
Feb 27 to March 13th CHIRPS-GEFS 15 day rainfall
Forecast anomalies (https://www.chc.ucsb.edu/data/chirps-gefs)
Accessed February March 2nd

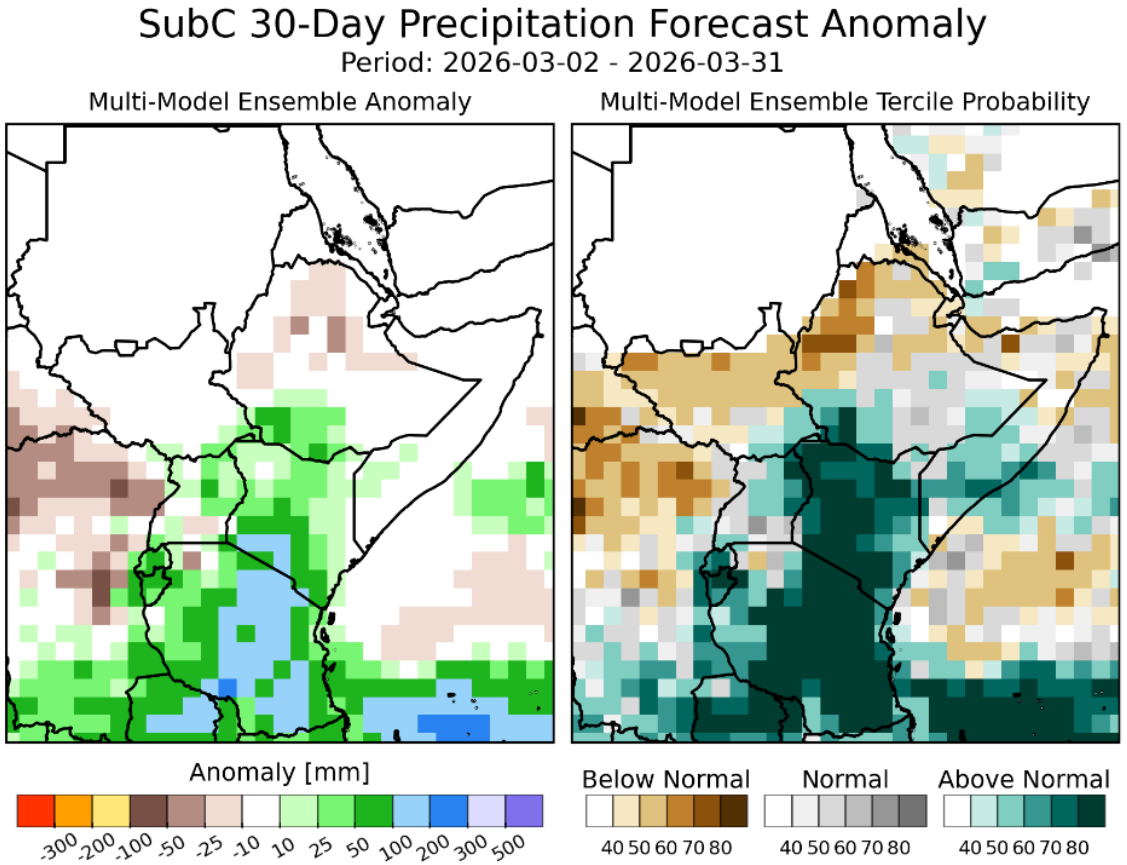
ECMWF Forecast from KIT/KMD Collaboration



https://github.com/alecjong-lab/ECMWF-S2S4AFRICA/blob/main/plots/Kenya/2026-02-28/dekadal/dekadal_precip.png

CHC SubC Forecast Ensemble

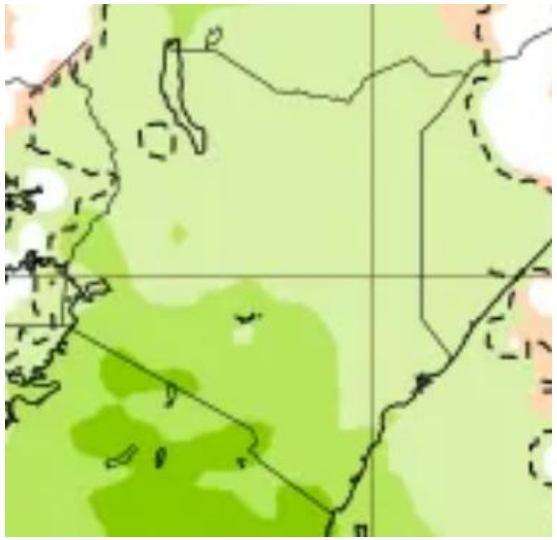




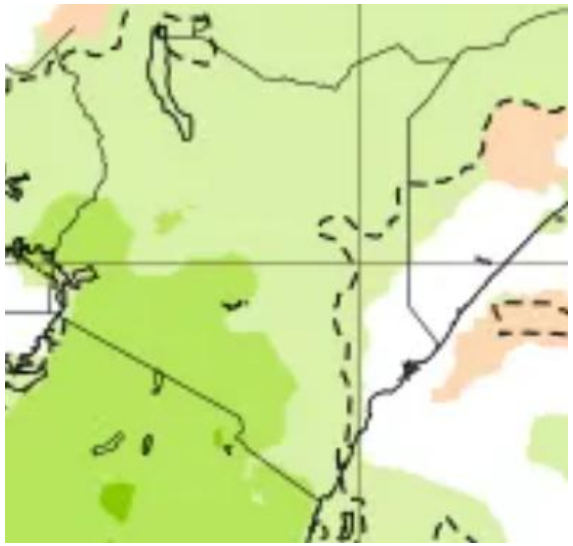
https://chc.ucsb.edu/monitoring/subx

ECMWF weekly forecasts are all above normal

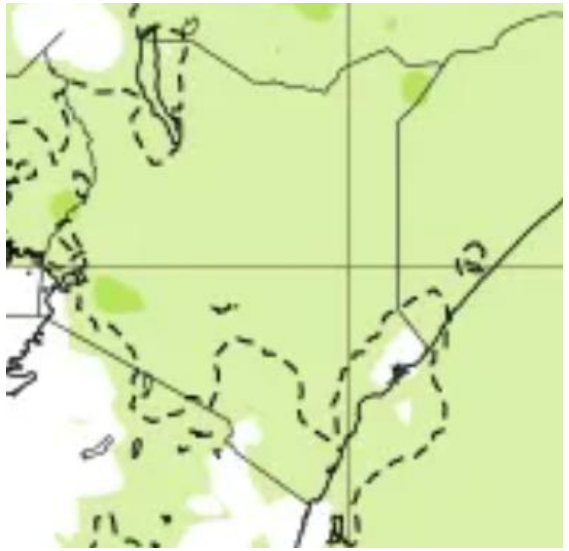
March 9-16



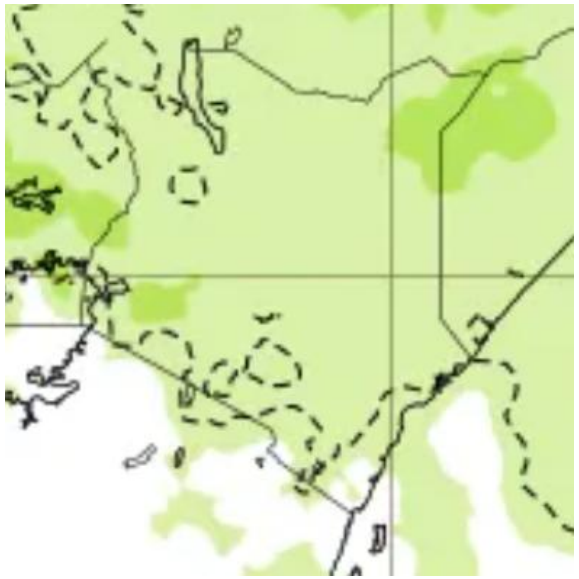
March 16-23

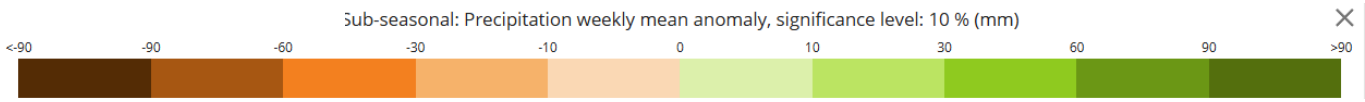


March 23-30



March 30th - April 6th

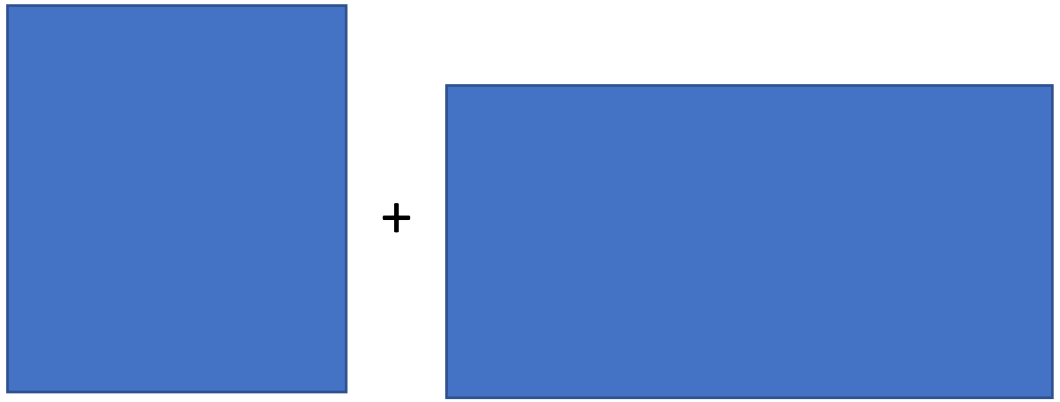




Question: **Where do ECMWF S2S rainfall forecasts suggest that a Start of Season is Likely?**

'Start of Season' is defined here as a dekad with at least 25 mm of rain, followed by 2 dekads receiving at least 20 mm of rain in total

We are working with ECMWF rainfall forecasts, downscaled to 0.05° using CHIRPS3 and quantile matching



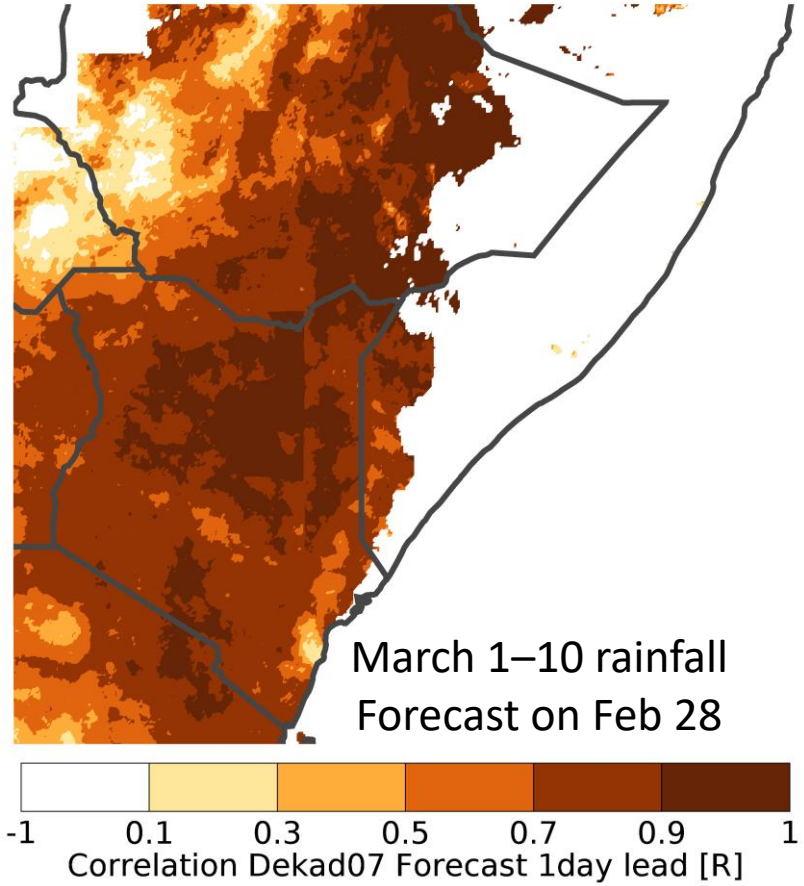
March 1–10 rainfall
> 25 mm

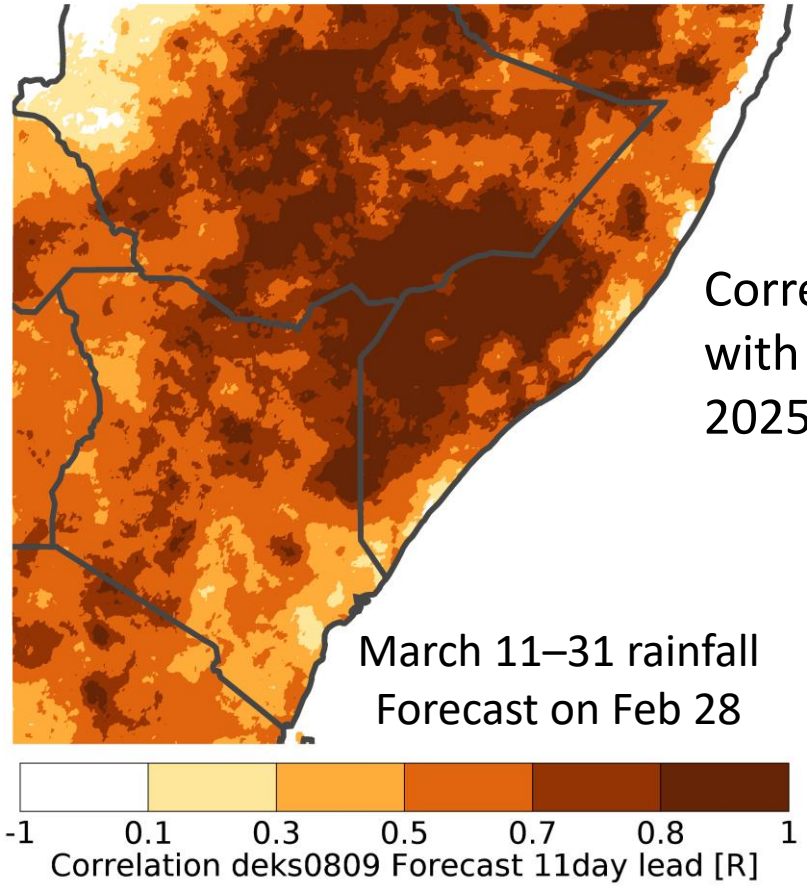
March 11–31 rainfall
> 20 mm

➔ Start of Season =
Enough Moisture
For Seeds to Germinate

Correlation Maps – ECMWF and CHIRPS3

Screened based on mean rainfall



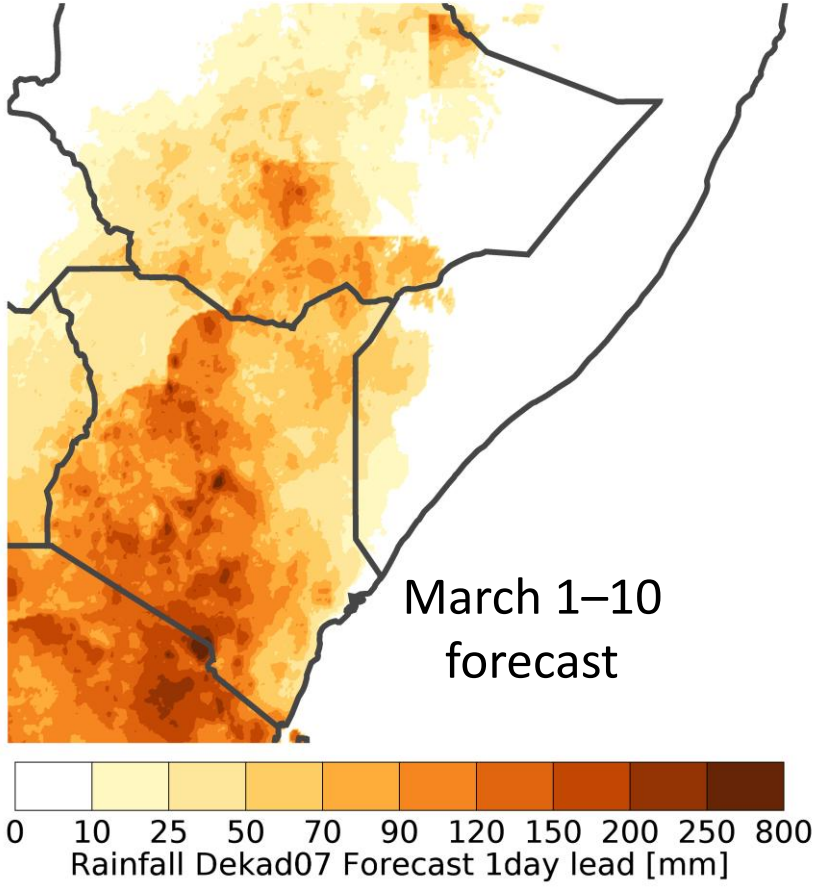


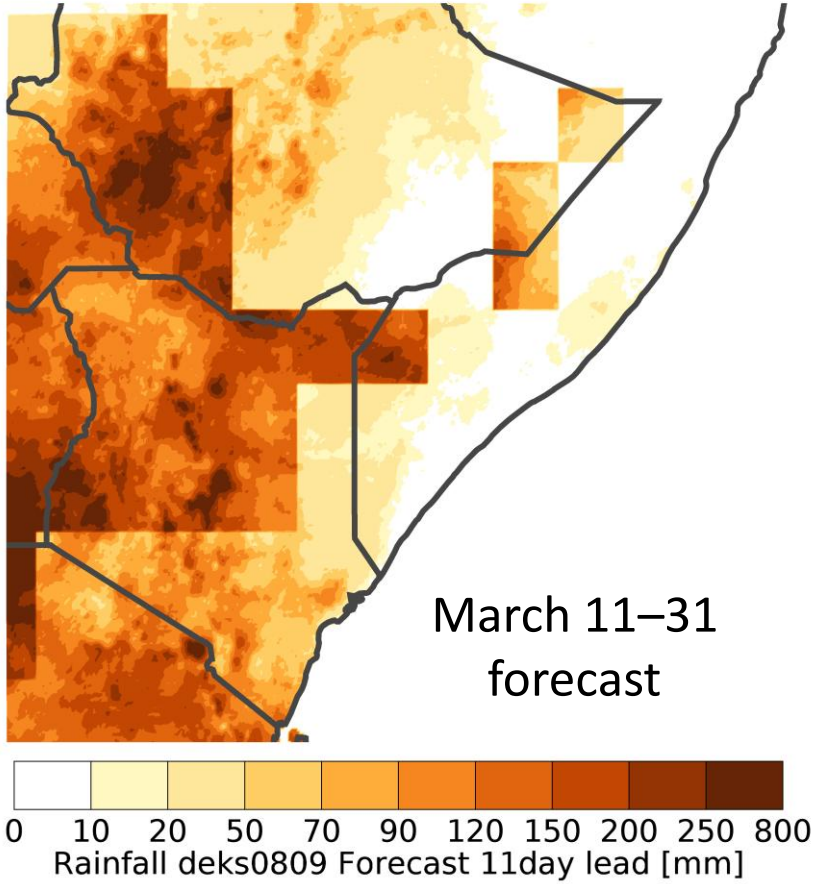
Correlation of forecasts with observed (2005-2025) rainfall

Analysis based on Feb 28th ECMWF forecasts

Forecasts from February 28th [mm]

Screened based on mean rainfall





Analysis based on Feb 28th ECMWF forecasts

Combining Forecasts and Standard Errors

- **Question**: Where do we think we will see 25 mm of rain in the first dekad of March or at least 20 mm in the last two dekads of March?

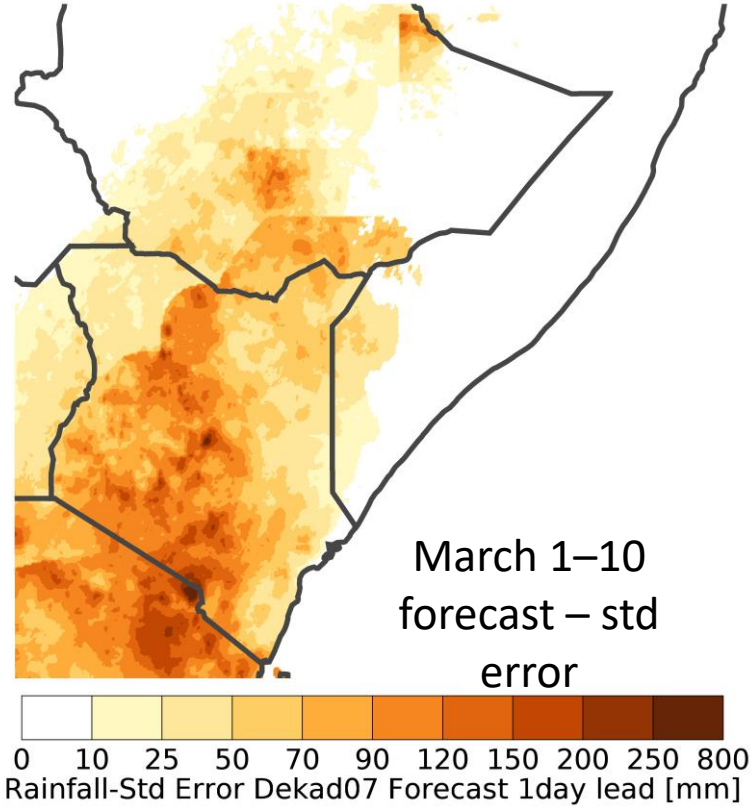
- Rainfall forecast minus 1 standard error seems a good lower confidence bound for the forecasts

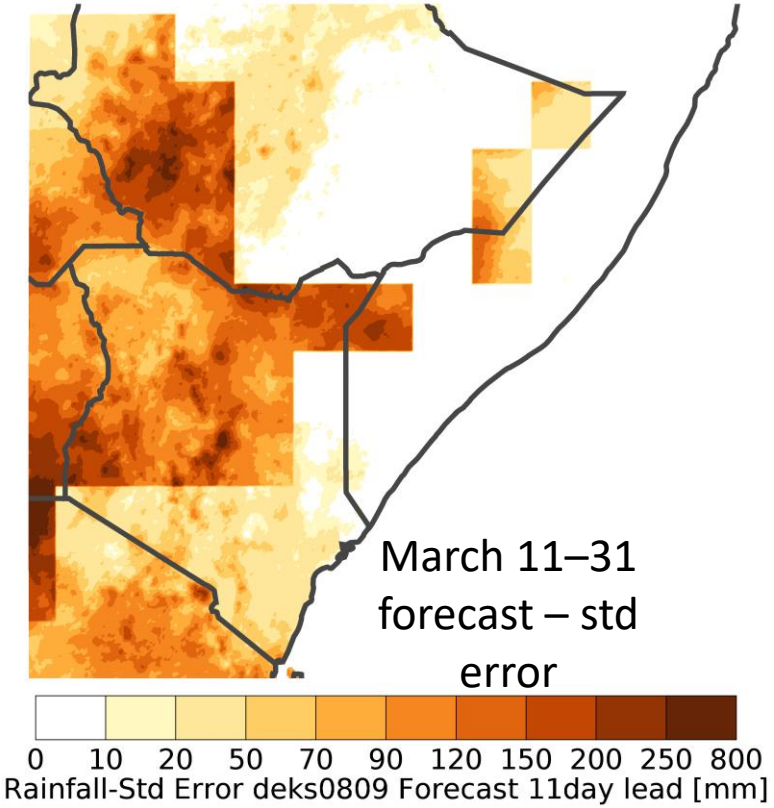
- There should be about an 84% chance of getting at least this much rainfall, at each 0.05° grid cell assuming errors follow a normal distribution

Forecast Minus Standard Error Estimate [mm]

Screened based on mean rainfall

- These maps show, in mm, a reasonable minimum expectation for rainfall totals in the first dekad of March and the last two dekads of March. These amounts are based on the February 28th ECMWF forecast and historical 2005–2025 standard errors, based on comparisons with observations.

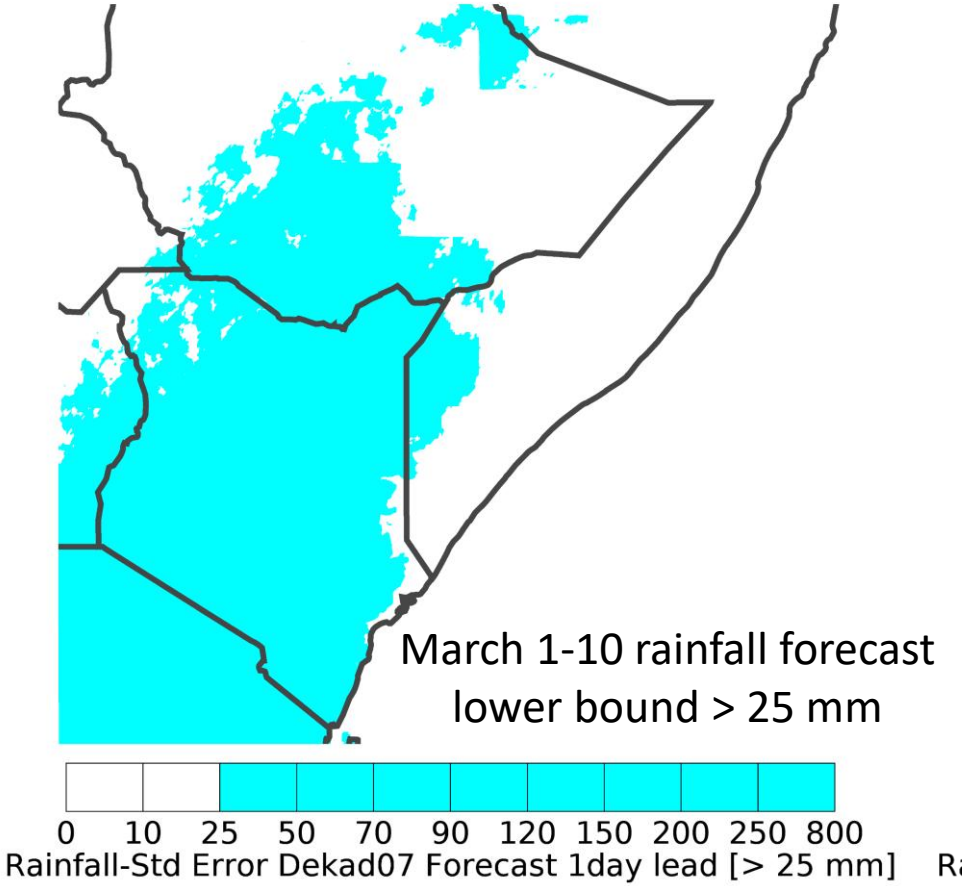


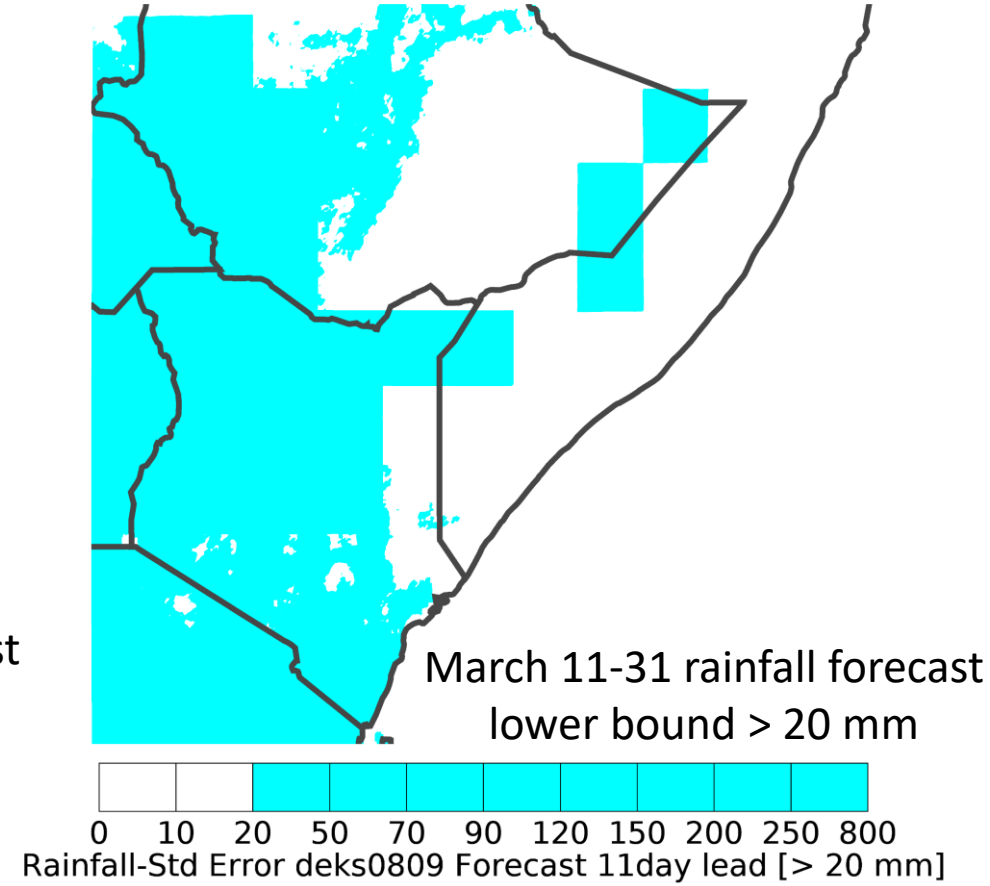


Forecasts based on February 28th ECMWF forecasts

Forecast suggests high probability of getting at least this much rain

Screened based on mean rainfall



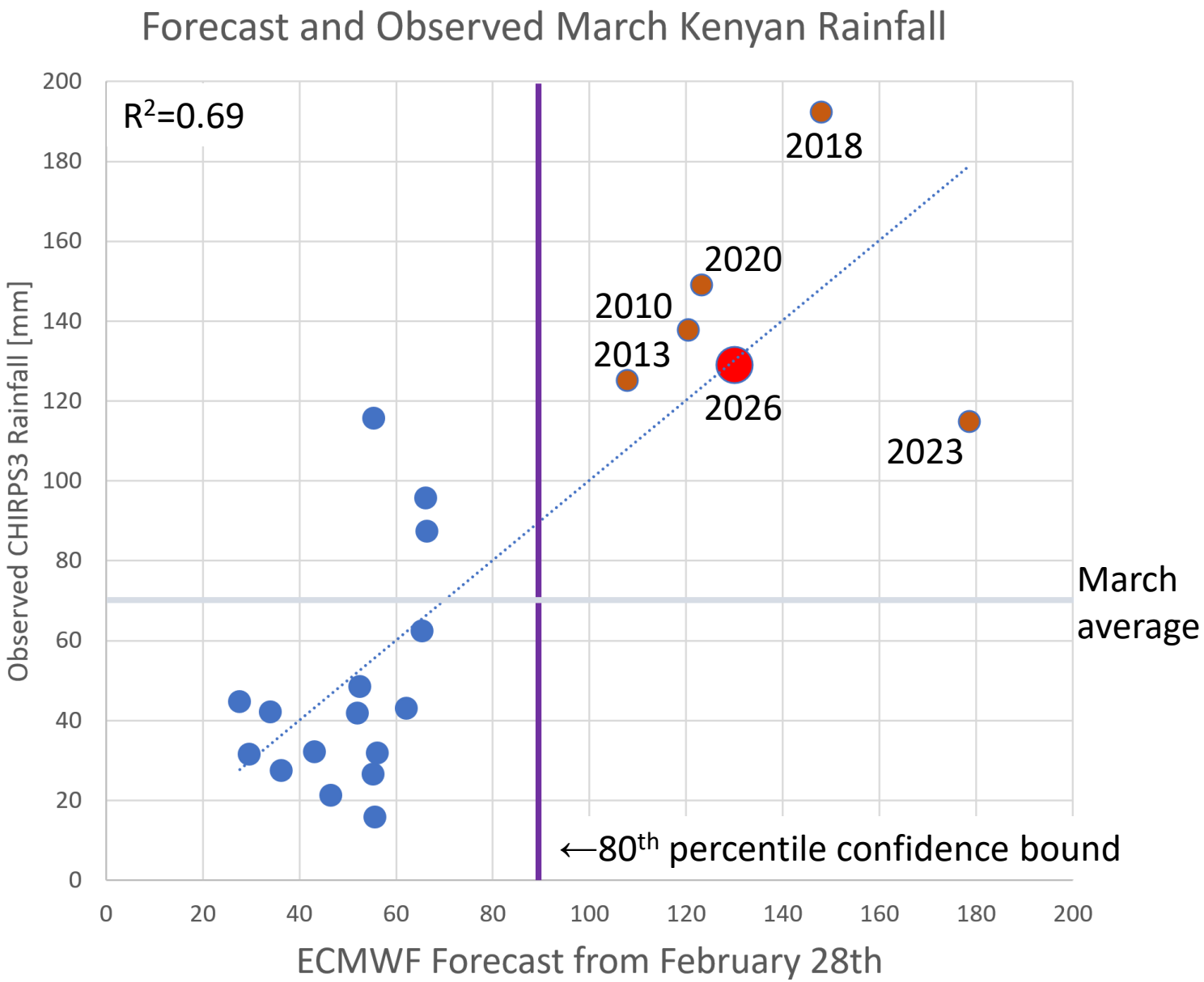


Main Result:
Almost all of Kenya
(cyan regions) will likely satisfy Start of Season criteria =
25 mm or more in March 1–10 &
At least 20 mm between March 11–31

Analysis based on Feb 28th ECMWF forecasts

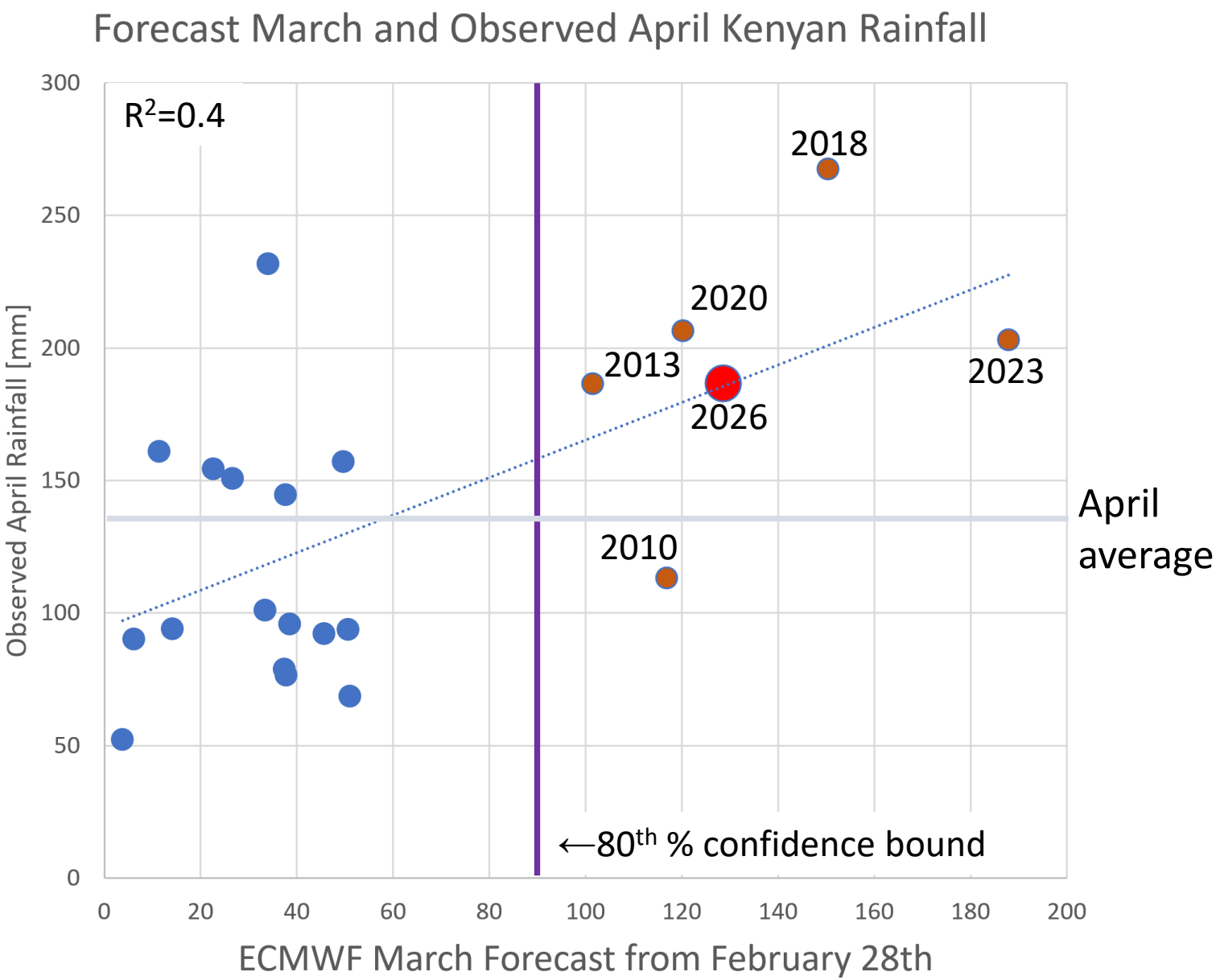
National March Forecast Looks Very Wet

- This plot shows 2005–2026 ECMWF forecasts for March for all of Kenya plotted against observed March rainfall.
- There is high forecast skill (R^2=0.7). The 2026 forecast is very wet. When similar forecasts were made in the past, the observed totals were very wet (2010, 2013, 2018,2020, 2023).
- The 2026 observed value has been set equal to the forecast.
- The vertical purple bar shows the 80th percentile confidence bound.

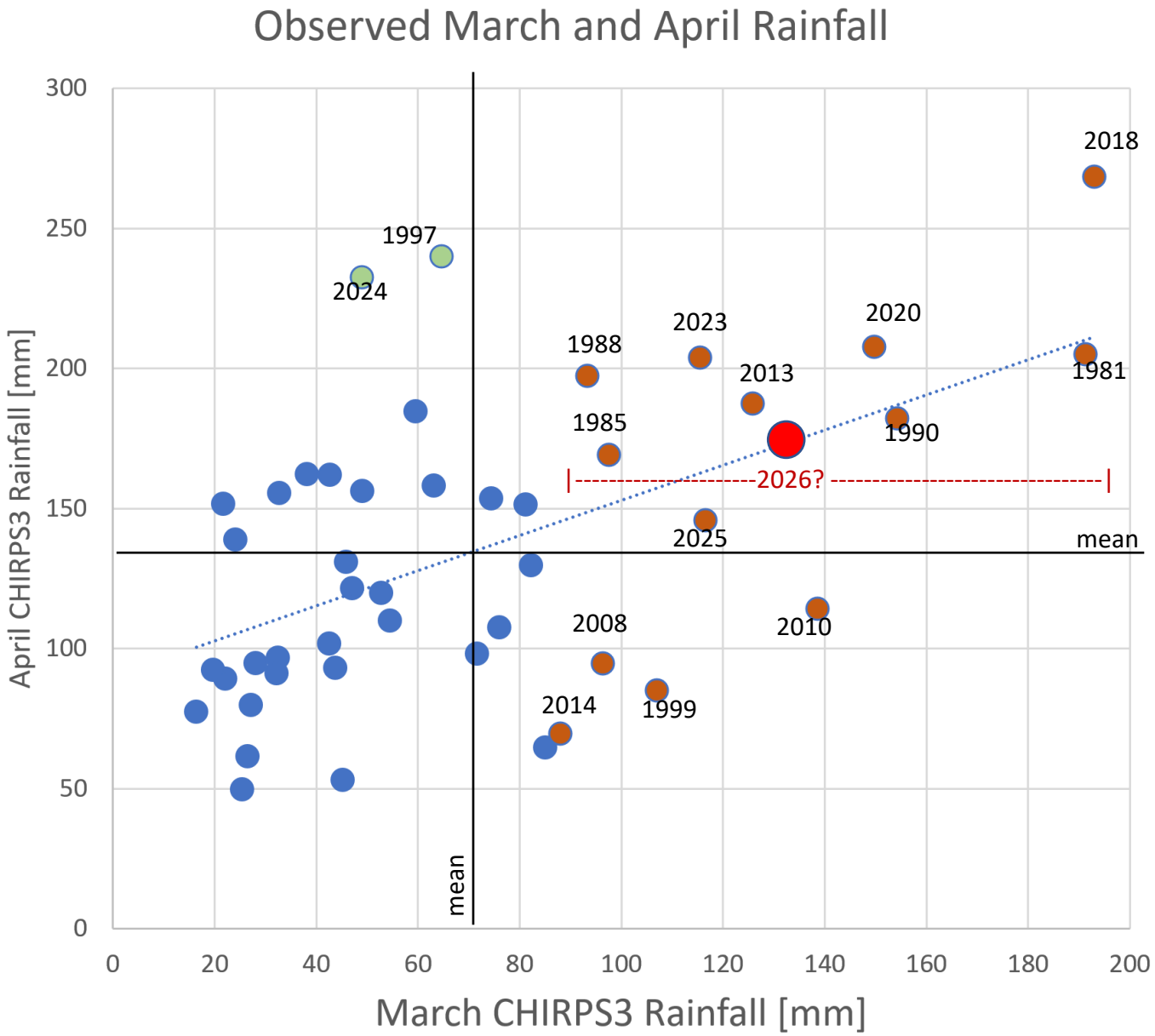


When March Forecast was wet Observed April rains were wet

- This plot shows 2005-2026 ECMWF forecasts for March for all of Kenya plotted against observed **April** rainfall.
- There is a reasonable relationship (R^2=0.4).
- When similar forecasts were made in the past, the observed totals were very wet in 2013, 2018, 2020, 2023 but not 2010.
- The 2026 observed value has been derived via regression.
- The vertical purple bar shows the 80th percentile confidence bound for the March forecast.

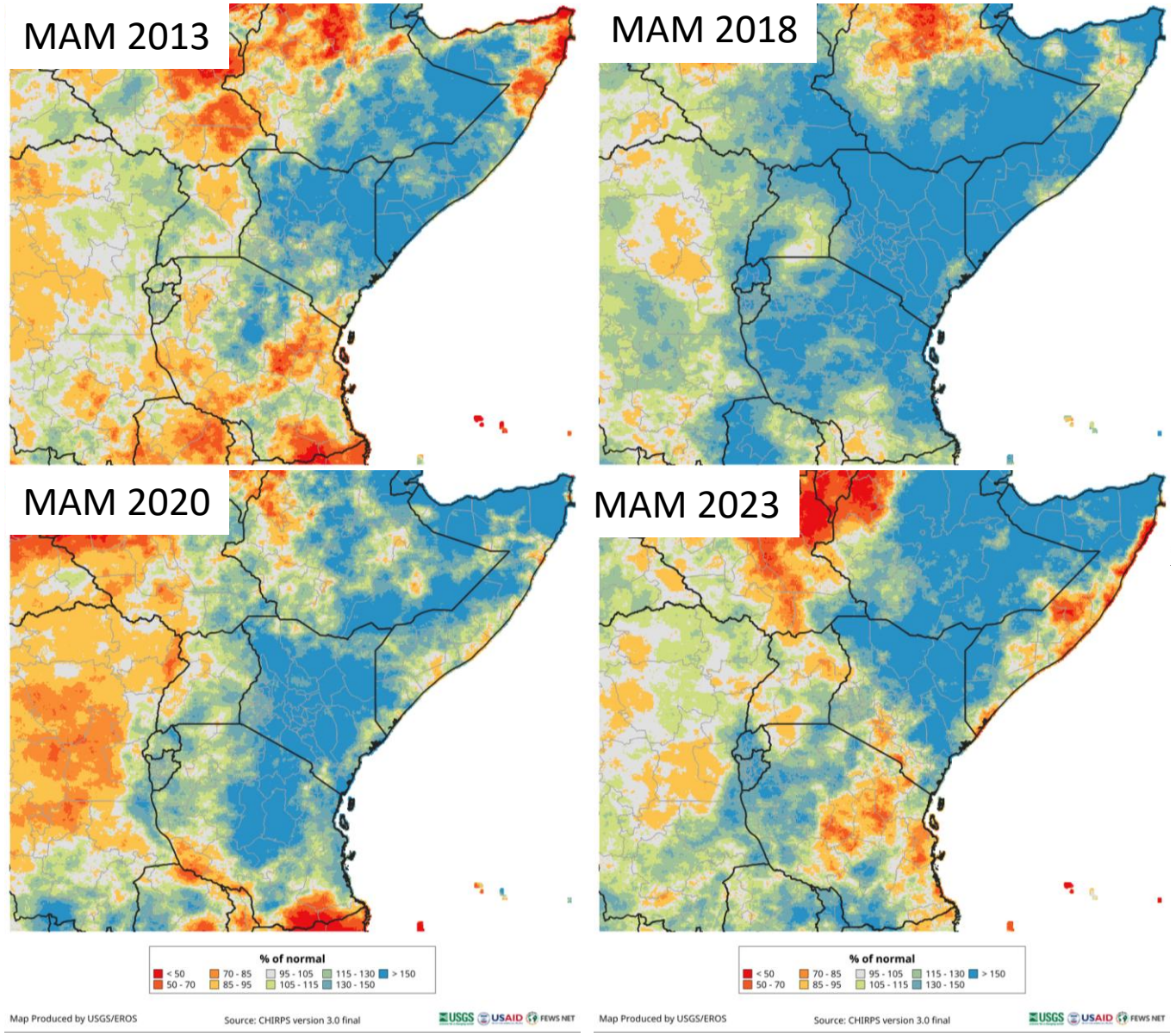


Defining A Larger Set of Analogs Based on Observed Wet March Months



Analogs: 1981, 1985, 1988, 1990, 1999, 2008, 2010, 2013, 2014, 2018, 2020, 2023, 2025

We might be looking at a real opportunity for good harvests
MAM CHIRPS Percent Anomalies



Conclusions

- After a very poor 2025 short rains, most of Kenya looks very likely to see an early start of the 2026 long rains season.
- ECMWF, GEFS and the SubC forecast systems all seem to converge on a similar positive outlook.
- The February 28th ECMWF forecast suggests that wet conditions in the first dekad of March will be followed by at least 20 mm of rain during the last two dekads of March – satisfying the Start of Season criteria.
- National averages of observed Kenyan rainfall suggest a strong relationship between wet March and wet April outcomes. Based on the observed relationship, if March rainfall is above-average, then April is usually also above average. This has occurred in 9 out of 13 analog years.